Products & Descriptions

Solenoid Valves
We offer direct acting 2 position 3 port solenoid actuated valves for use as pilot valves and for single stage operation.
A number of solenoid thrusters are available, including industrial enclosures, subsea designs and ATEX approved solenoids for use in potentially explosive hazardous areas. Manufactured mostly from stainless steel, valves are available as standard ported designs or for manifold mounting, and for use with a range of media, including air and other gases, oil, water, water glycol mixes and sea water.

Stop & Metering Valves
Our range of stop and metering valves can be supplied to suit your installation configuration requirements, including inline, manifold mounted or cartridge types for inclusion in your own manifold. Stop valves provide a full positive shut-off function, while metering valves, which are mostly supplied with a graduated micrometer style adjustment, can be used to provide speed control or any other simple orifice function where adjustment is required. Pressure compensated flow control valves are also available, providing a constant flow rate in situations where the load or supply pressure varies.

3 Ported Directional Control Valves
The 3B series of 2 position 3 ported directional control valves are available for use on a wide range of media, from gases to sea water and many difficult and hazardous chemicals and fluids. Using a balanced poppet design, actuation of these valves is completely independent of the main line pressures, making switching reliable and consistent. With a vent chamber between the main ports and the pilot system, different pilot and main line fluids may be used without risk of accidental mixing or contamination. Various actuation methods are available, including hydraulic and pneumatic pilots, levers and cams.

4 Ported Directional Control Valves
The BiS range of 4 port directional valves includes both balanced poppet and rotary shear seal designs, commonly used to give control of double acting actuators. The '4R' series valves are lever operated 2 or 3 position rotary valves offering 90° manual actuation, in both detented and non-detented configurations, and with a range of centre conditions. They may be used in many different installations, with standard threaded porting, directly mounted on manifolds, or mounted behind panels. When manual control is not required, the '6B' range of 2 position 4 way valves offer other methods of actuation, such as hydraulic and pneumatic pilots.

Check and Shuttle Valves
The BiS range of inline check valves is a comprehensive range comprising nominal bore sizes from 1/4" to 2". Manufactured from stainless steel, they are suited to use with a wide range of liquids and gases, with pressures up to 1379 bar (20,000 psi). High cracking pressure check valves are also available by contacting the factory. In addition, there are cartridge check valves for use in manifolds or for direct integration into other equipment. The 'SV' range of shuttle valves may be selected when it is necessary for a valve to select the higher of two pressures, a function often used when redundancy of pilot systems is required.

Pilot Operated Check Valves
Our standard range comprises single and dual pilot operated check valves, decompression checks designed to reduce shock when a high pressure line is opened, and unloader valves. Actuation may be achieved by hydraulic or pneumatic pilot, lever or cam.

Relief Valves
We offer a comprehensive range of safety relief and relief valves for use on a wide range of media, from low pressure valves that vent to atmosphere, to high pressure (1379 bar, 20,000 psi) piped return valves. Typically, low pressure valves offer a pop-open style characteristic, while high pressure valves offer proportional or metering style operation.
BiS relief valves are CE marked to denote approval as Category IV safety accessories under the 97/23/EC Pressure Equipment Directive. Many options are available, including cartridge and manifold mounted versions, and many valves are available in exotic materials including monel, inconel, hastelloy and titanium.

Excess Flow Valves
Our range of excess flow valves are designed to meet the exacting standards of the oil & gas and petro-chemical markets. Their primary function is to shut a line down in the event of breakage or rupture of a downstream pipe or hose, or any other application where excessive flow must be avoided.

Filters
BiS Valves offer a wide range of filtration products, ranging from small component protection filters or strainers to full pressure main line filter units. Available with a wide range of mesh ratings, our F and EF series filters can provide vital protection for contaminant sensitive downstream components, and are particularly designed to protect against rogue pieces of debris that have passed the main line filtration or were built into the system during manufacture. Our SF and DF filters may be used as main line filters in the supply line, and offer various features such as bypasses, clogging indicators, and for the duplex style filters, on-line servicing.

Pressure Sensing Valves
The '3SV' series-sensing valve is configured to provide a low-pressure signal switch, dependent on a separate sensed pressure of up to 414 bar (6,000 psi). The product can be used to provide a pilot pressure signal to actuate another valve when the status of a second service rises or falls beyond a pre-set pressure.

Pumps
Our range of low flow piston pumps offer high pressures up to 827 bar (12,000 psi). Designed for use primarily with mineral oil, certain limited use with water glycols is also permitted.

Design, Manufacturing, Testing & Quality

Design

All BiS Valves products are supported by a team of hydraulic design engineers. As well as making continuous improvements to the current range and designing new products, the team is ready to help you find the right solution for your application.

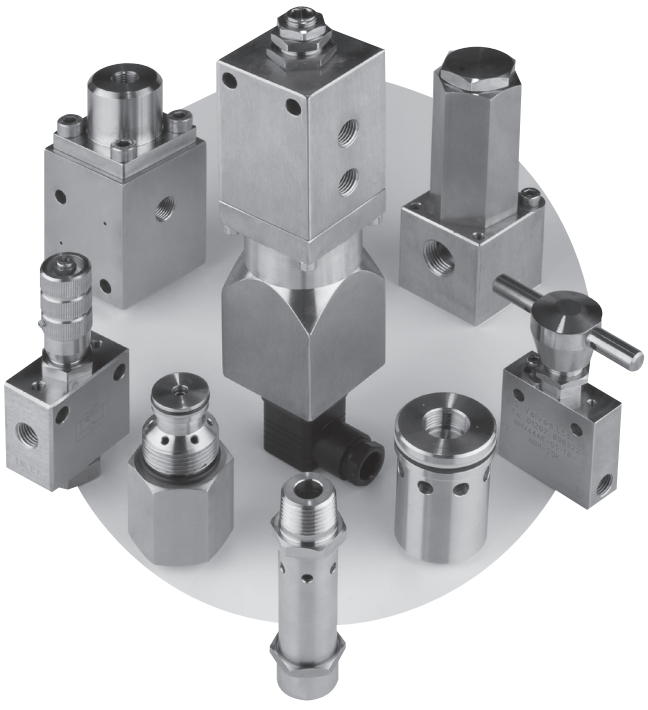
Our design service can include making simple modifications to existing valves to make them an exact match for your requirements, through to development of a complete new valve.

Among our most popular product solutions are hydraulic manifolds where we can incorporate several valves into a single unit. This improves installation, reducing the number of expensive fittings required, makes servicing easier since pipe work does not have to be removed when a valve needs servicing, and minimises potential leakage points in the system.

We can also offer design consultancy services and we have the capability to work with you to design new products and systems, and to develop your existing designs. Contact us for details.

Manufacturing

Manufacturing of BiS Valves products is mostly carried out in our own factory near Wimborne, Dorset, where we have extensive machining facilities devoted to close tolerance manufacture of stainless steel valve parts. In order to achieve the high levels of seat integrity required by the demanding low leakage applications that we serve in the offshore oil and gas industry, sound manufacturing and assembly techniques are essential, and are even more critical when handling more exotic materials such as monel, inconel, hastelloy and titanium.



Testing

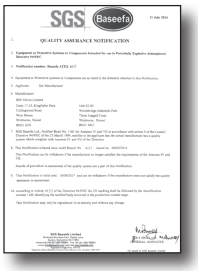
BiS Valves has an extensive and growing list of in-house testing capability, both for production testing and for qualification work.

This includes hydraulic testing to 2,069 bar (30,000 psi), flow and pressure drop testing,pressure decay and leakage testing. Test mediums include mineral oil, water, various water glycols and nitrogen. Our capabilities, which are augmented by external testing when necessary, also include high pressure gas testing, environmental testing, hyperbaric facilities and shock and vibration testing.


BIS VALVES LIMITED

BS EN ISO 9001:2008

Scope of certification


SGS Baseefa

QUALITY ASSURANCE NOTIFICATION


CERTIFICATE OF QUALITY SYSTEM APPROVAL

Quality

As an ISO 9001:2008 company, our objective is to be your preferred partner and supplier, providing you with product solutions that are engineered to meet your requirements, delivered on time, with a commitment to through-life support and above all, to the quality standards demanded in today's business arena. We will continue to achieve this by focusing all personnel into sustaining the standards established by our quality system and through continuous improvement and innovation in our products, design, manufacturing and business systems.

At BiS Valves quality is of paramount importance and particular care is taken to ensure all products meet the high standards required and are tested to ultimately ensure compliance.

Our Quality Assurance Management System approvals include:-

- BS EN ISO 9001:2008 - Quality Systems
- ATEX Directive 2014/34/EU
  Equipment for use in potentially explosive atmospheres
- Pressure Equipment Directive 2014/68/EU (PED)

FPAL empowered by Achilles